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2019, ICRTAC 2019Sketch based image retrieval using grid approach on large
scale databaseG.G Rajput^{1*}, Prashantha²¹ Akkamahadevi Women's University, Vijayapura 586108, Karnataka, India² Rani Channamma University, Belagavi 591156, Karnataka, India

Abstract

Several approaches are found in study for image retrieval from the database. A sketch based system is an efficient method which does not require the user to have a high skill to draw the query sketch. Sketch-to-image matching system on a large-scale image database should ensure that the system always tries to find correct matches for a given sketch query. In this paper, we present an efficient method for image retrieval from large database using sketch as a query image. Features are extracted from the image by placing a non-overlapping grid and overlapping grid over the image, respectively. Using a weighted similarity approach, the most similar images to the query are then retrieved from the image dataset. For performing experiments images of flickr15k dataset are used. The proposed method shows excellent retrieval performance in comparison to methods available in the literature.

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1. Introduction

In recent years, images are captured in digital form and stored in multimedia database and there is an increased

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demand for powerful and efficient retrieval system for browsing, searching and retrieving images. Image search requires user to supply query to the system in terms of keyword or image file/link, or click on some image; the system will return images relevant to the query. The similarity used for search criteria could be meta tags in terms of annotations, color distribution in images, region attributes, shape attributes, etc. A Metadata search criterion relies on annotation quality and completeness. Further, the process can be time consuming and that keywords desired to describe the image are not adequate. During retrieval process, keyword based image search is subjective and has not been well-defined. Textual based search becomes a time consuming task for very large databases or for the images that are generated continuously from the devices such as surveillance cameras. During the search process, many images may be missed that use different synonyms in their descriptions.

Content Based Image Retrieval (CBIR) plays a major role in retrieval system, and is widely studied recent years [1, 2]. CBIR avoids the use of textual descriptions. Images are retrieved from the database based on similarities in the image contents (colors, shapes, textures, etc.) between user-supplied query image or features of the query image and images from the database. An image is retrieved in CBIR system by adopting several techniques [3]. However, due to the lacking of universal features extraction and effective database indexing, its works efficiently on only small-scale image database. Humans look for images using other images, and generally follow this approach for categorizing the images. One of the efficient methods under CBIR for image retrieval is Sketch Based Image Retrieval. Sketch Based Image retrieval (SBIR) conveniently sits between text-based image retrieval (TBIR) and CBIR. In SBIR system, sketch content (in the form of freehand sketches) is input to retrieve images whose shapes match the sketch provided by the user. Rapidly changing the way of user input query in the form of freehand sketches due to the growing touch-screen technology, with non-availability of appropriate and complete set of tags, sketch-based image retrieval techniques become vital for retrieving images from large databases. In this paper, our purpose is to develop a sketch based image retrieval system which can retrieve the digital images from large database using free hand sketches as query.

In SBIR, many studies have been focused on extracting descriptors. Generally, sketches are divided into the blocks and the descriptors such as color, texture and shape are determined for matching the images. After matching, the system will retrieve the related images stored in database. Global features like Edge histogram descriptor (EHD) [12], Histogram of gradients (HOG) [5], and Angular Radial Partitioning (ARP) [6] are widely used for image analysis and classification.

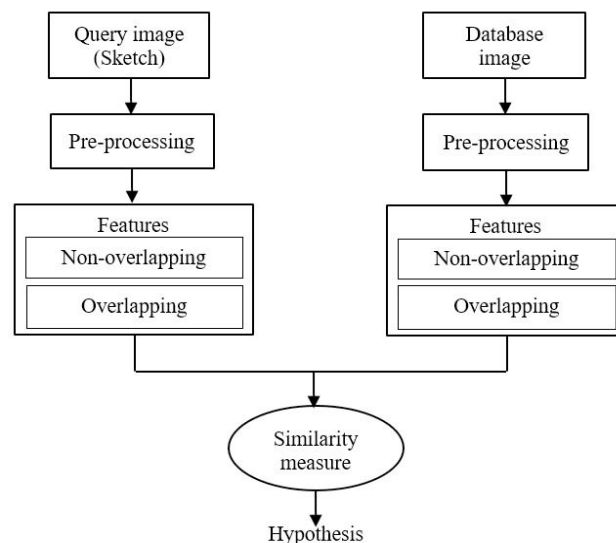


Fig. 1. Block diagram of Sketch Based Image Retrieval

In this paper we propose an efficient method for retrieving images from large database based on sketch as query using grid(block) based feature extraction. Global contour of the sketch/image is divided into non-overlapping blocks and features are computed for each block (F1). Next, the sketch/image is divided into overlapping blocks and

features are computed for each of block (F2). Weights are assigned to F1 and F2 empirically. The matching process retrieves similar images from the database. The block diagram of the proposed approach is shown in Fig. 1.

Rest of the paper is described as below. Section 2 presents review of literature; database description and pre-processing in section 3. Section 4 proposed methodologies, section 5 Experimental evaluation and section 6 contain conclusion.

2. Review of literature

Comparing the sketch to all images in the database directly is impractical. Instead, descriptors are extracted and stored for fast access. The descriptors can be represented as points in a high-dimensional space and close match is found by searching nearest neighbours in this space. Many methods have been proposed in literature to extract descriptors, global or local. In this section, we briefly describe few approaches which are widely used in SBIR system in recent years. The difficult task in SBIR system is matching the rough sketch of feature lines to natural image. The first approach is addressing this problem is presented by N. Chang [10]; searching the natural image based on pictorial description. The SBIR systems are used in the criminal investigation for identification of unsubstantial images. Similar applications are implemented in [8], [9]. Hirata et al.[7] dataset contains 205 colored oil-painting images and its search the matching sketch image edges against the database image edges. Images are subdivided into 8×8 local blocks after the normalization process, for each local block, the best local correlation is computed by searching in a small window of local blocks. In the work presented by Rui Hu et al. [4], Multi scale region segmentation with a bag of visual words framework is used to retrieve image given sketch as query. Image is decomposed into region representations at multiple scales and regions are collected from the nodes of a hierarchical region tree. Then, contours of the region map are extracted. The contours essentially capture various levels of details. Local structures are captured using multi scale HOG descriptor built on the dense gradient field (GF-HOG). Bag of Visual Words (BoVW) codebook is formed using k-means and histogram is generated for matching. GF-HOG approach is used to encode the query sketch. Finally, the sketched object is localized in the retrieved image.

In the work presented by Eitz et al. [12], grids are used to locate photos using sketched description of object shape. Descriptors from each cell are concatenated to form a global image feature which offers limited invariance to changes in position scale or orientation. The images are divided into grids, and the color and texture features were determined in these grids. The grid based approaches are also found in other algorithms, for example in the edge histogram descriptor (EHD) method Eitz et al. [13]. Szanto et al.[15] proposed user interface system for retrieval system, there are three descriptor are used like Histogram of oriented Gradients(HOG),Edge Histogram Descriptor(EHD), Scale Invariant feature Transform (SIFT)to extract the feature of the original image and Minkowski distance has been used to find the distance between original image and sketch image. This retrieval system has been implemented on small dataset. Jain et al.[14] proposed linear combination of color histogram and edge histogram similarity measures and combined color and shape information to retrieve the trademark images from the database of 400. In this paper, sketch based image retrieval system is proposed for retrieving images from a collection of flickr15kdataset given a free-hand sketch as query.

3. Database description and pre-processing

The proposed method performed on flickr15k dataset which is created by Hu [17]. The detail description of dataset and pre-processing are presented below.

3.1 Database description

We evaluate our proposed approach using flickr15k dataset [17]. The flickr15k dataset contains 10 classes of sketch and 60 classes of natural images. Each sketch class of contains 33 sketch images, the total number of sketch

images are $10 \times 33 = 330$, and total number of natural images from 60 classes is 15k, like eiffel_tower, bird and Egypt_pyramidetc , sample images from the dataset (natural and sketch) is shown in fig.2.

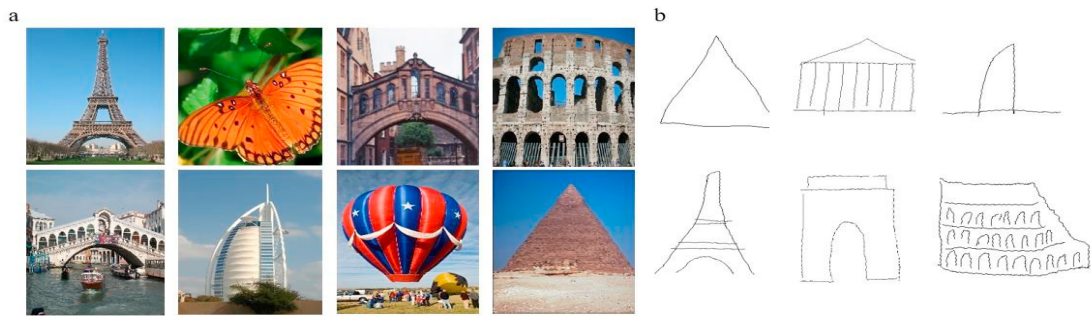


Fig.2. Example of dataset images (a) natural images; (b) sketch images

3.2 Pre-processing

In the pre-processing stage, we extract global information from the natural images. Otsu's method [18] is used to obtain the contour of the natural images. Subsequently weak contours are eliminated. Sample natural images and contours of the images obtained by applying Otsu's method is shown in fig.3.

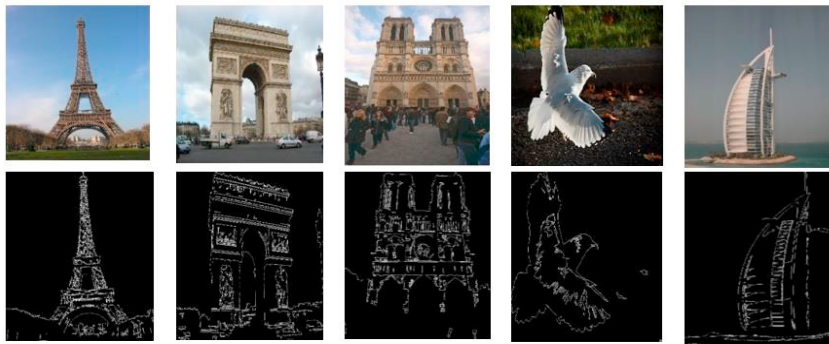


Fig.3.Original images in first row and global contour images in second row

4. Proposed methodology

We propose a two way feature extraction process for retrieval of images given a free-hand sketch as query. The details of the methodologies are presented below.

4.1 Non-overlapping grid based image retrieval

The global contour is resized to 260×260 pixels, the size being determined empirically. The contour is then divided into 100 sub images by placing a non-overlapping grid of size 10×10 shown in Fig.4. For each sub image, mean of the pixels values is computed. All the values are then concatenated to form a feature vector. Given a input sketch, the sketch is resized to 260×260 pixels; divided into 100 sub images as described above and feature vector of mean values of sub images is obtained Euclidean distance between the feature vector of the sketch mage and feature vectored of the picture stored prior are computed. The resulting images in the order of ascending distance values

computed are then retrieved from the collection.

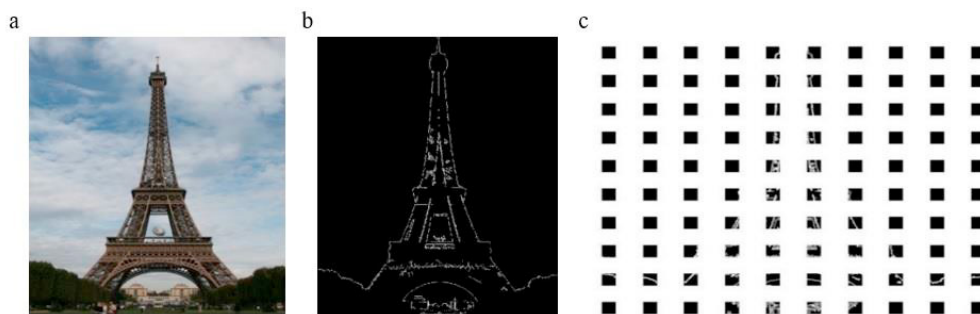


Fig.4. Feature extraction using non-overlapping grid approach (a) Original image; (b) global contour; (c) feature extraction

The proposed method result presents in two steps; first one is out of 10 resulting images, 8 images are matched to the user query and remaining two is most similar to the user query. In second approach, out of 20 images retrieved, 17 images are matched to the user query and remaining images are mismatched fig.5.

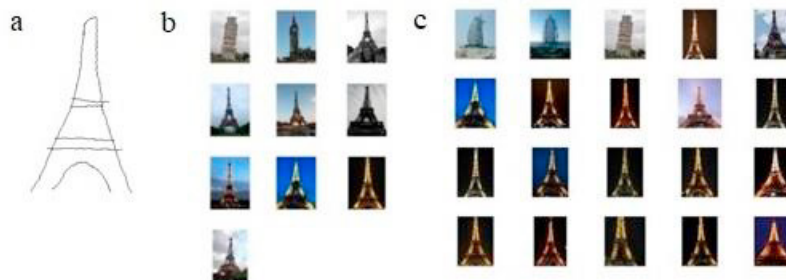


Fig.5. Non-overlapping grid results (a) sketch image; (b) top ten results; (c) top twenty results

4.2 Overlapping grid based image retrieval

In order to retrieve the most similar images from large database we proposed another method: Non-overlapping grid based image retrieval; the global contour image is divided into 100 sub images, each sub image overlapped by 20% in two side (left side and upper side) Fig.6. Rest of the procedure defined in above section is followed for image retrieval given a sketch as query.

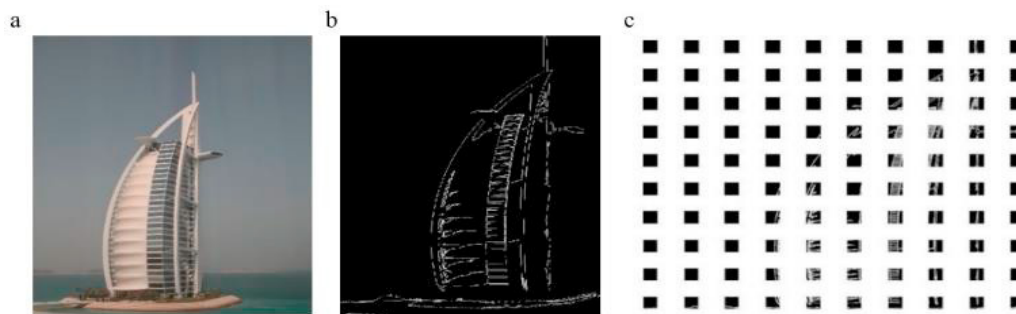


Fig 6: Feature extraction using overlapping grid approach. (a) Original image; (b) global contour; (c) feature extraction

The overlapping approach result presents in two steps; first one is out of 10 resulting images 8 are correct and 2 are mismatched results; and inn second approach, 18 out of first 20 images are correctly retrieved Fig.7.

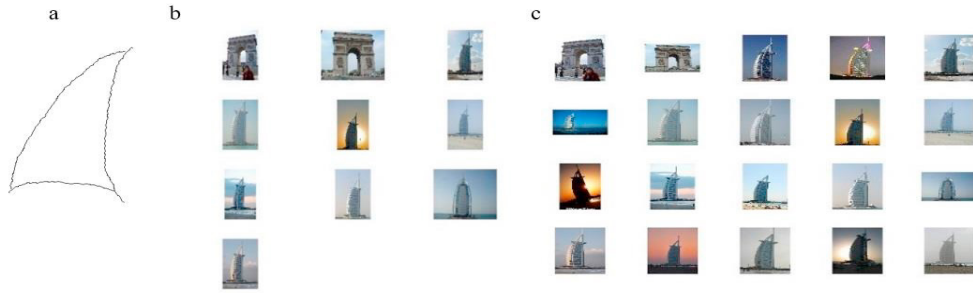


Fig.7. Overlapping grid results (a) sketch image; (b) top ten results; (c) top twenty results

4.3 Weighted approach for image retrieval

In weighted approach, the feature vectors computed in section 4.1 and section 4.2 by using non-overlapping grid and overlapping grid over the contour map, respectively, are combined by assigning suitable weights to each of the vectors in the range $[0,1]$ [16]. The procedure is defined below.

The feature set is denoted S_n^m where n denotes the number of dataset images and m denotes feature vector. We denote S_n^1 as feature vector obtained using non-overlapping grid approach and S_n^2 denotes feature vector obtained using overlapping grid approach. Given a sketch as query, let S^m denote the feature vector of the sketch; S^1 denoting features obtained using non-overlapping grid approach and S^2 denoting feature vector obtained using overlapping grid approach. Let $\text{hyp1}(n)$ and $\text{hyp2}(n)$ denotes the similarity between S_n^1 and S^1 and similarity between S_n^2 and S^1 , respectively. Then similarity of sketch and database image by taking non-overlapping grid and overlapping grid similarity as follows:

$$\text{hyp}(n) = w * \text{hyp1}(n) + (1 - w) * \text{hyp2}(n), \quad n = 1, 2, \dots, N \quad (1)$$

Let w denote the weight assigned in the interval $w \in [0, 1]$ and $n=1,2,\dots,N$, where N denotes number of dataset images. We set $w=0.8$ in our experiment. Finally, to retrieve the top- n ranked we sort the similarity score in descending order. Given a sketch, most similar images are then retrieved using the weighted approach. In top 10 search results, 9 results are correctly matched and remaining one is also most similar to the user sketch. In top 20 search results, 19 images are correctly retrieved. The results for sample sketch are shown in fig. 8. The results are encouraging compared to individual approaches defined above.

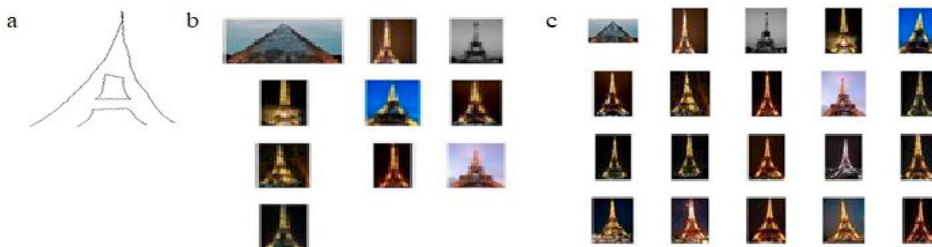


Fig.8. Results of weighted approach (a) sketch image; (b) top ten results; (c) top twenty results

5. Experimental Evaluation

The proposed method is evaluated on flicker dataset [17]. We have chosen 15k natural images and 330 sketch queries. Retrieval performance is evaluated using Mean average precision (MAP) [21].

$$MAP = \frac{1}{S} \sum_{s=1}^S AP(q), \quad AP = \frac{1}{R} \sum_{k=1}^N p(k) * rel(k) \quad (2)$$

Where S is number of Sketch's; R is number of related images in retrieved images; N is total number of retrieved images; p(k) represents top precision score; and rel(k) represents an indicator function, if rank r is relevant to the sketch query rel(r)=1, otherwise rel(r)=0. The proposed method is compared with other methods in the literature confining to state-of-art methods that are being evaluated using flicker dataset images. Table.1 shows the comparison of the proposed method with Bag-of-Visual-Words based methods such as HOG [17], GF-HOG [17], and Self Similarity (SSIM) [19] and also with non Bag-of-Visual-Words methods such as Structure Tensor [13], Shape Context [20].

Table.1. Comparison of proposed methods with methods in literature based on MAP

Methods	Classical approaches	MAP
Non-BoVW	Structure Tensor	0.076
	Shape Context	0.081
BoVW	HOG	0.111
	GF-HOG	0.122
	SSIM	0.095
Our Approach	Non-overlapping grid	0.129
	Overlapping grid	0.143
	Weighted based	0.151

The comparison reveals that our methods yielded better MAP values compared to MAP values of other methods on flickr15k database and that performance of weighted base approach is better with MAP value of 0.151.

6. Conclusion

This paper presented efficient approaches for SBIR system on large scale dataset. Features are extracted using non-overlapping and overlapping grid approaches. A key characteristic of our method is estimation of similarity between the features of sketch query and features of database images using weighted approach. The proposed method is experimented on images of flickr15k dataset. The proposed weighted based approach yielded encouraging results compared to other methods in the literature; the method retrieved the most similar images from chaotic dataset as compare to the results of other methods in the literature.

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